Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/23/2024
PI Name:	Shirazi-Fard, Yasaman Ph.D.		
Project Title:	Rodent Research Standard Housing Mission: Multidisciplinary Approach to Understanding Spaceflight Responses		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/01/2022	**End Date:**	04/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per NASA Ames Research Center (ARC): Dr. April Ronca, the previous Principal Investigator, has retired from NASA. The new PI for this investigation is Yasaman Shirazi-Fard, Ph.D. Previously, Dr. Shirazi-Fard was a Co-Investigator (Co-I) on this project. NOTE: Dr. Ronca will still support the project, but as a Co-I now (Ed., 12/23/24).		
COI Name (Institution):	Globus, Ruth Ph.D. (NASA Ames Research Center) Fuller, Charles Ph.D. (University of California, Davis) Alberts, Jeffrey Ph.D. (Indiana University, Bloomington) Vitaterna, Martha Ph.D. (Northwestern University, Evanston) Bodine, Susan Ph.D. (University of Iowa) Delp, Michael Ph.D. (Florida State University) Pecaut, Michael Ph.D. (Loma Linda University) Galazka, Jonathan Ph.D. (NASA Ames Research Center) Willey, Jeffrey Ph.D. (Wake Forest University) Karatsoreos, Ilya Ph.D. (University of Massachusetts, Amherst)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	RR-26 is a NASA Space Biology (SB) Program funded study for the 2nd Rodent Habitat Flight Validation, following the addition of the Nest Box (NB) as the current solution to meet OCHMO-550 requirements and to provide adequate bedding substrate and/or structures for resting and sleeping of animals per The Guide to the Care and Use of Laboratory Animals (The Guide). The Flight Institutional Animal Care and Use Committee (FIACUC) and the science community share a common goal to advance the utility of the Rodent Research (RR) Habitat for conducting high quality, rodent spaceflight experiments. The NASA Chief Veterinarian and FIACUC are concerned that the RR Habitat without a Nest Box (NB), fails to meet animals' needs and the regulatory requirements to provide a dedicated structure that allows mice to sleep, nest, and thermoregulate. On the other hand, growing concerns have been raised in the RR flight investigator community (based on results obtained from some missions since 2017) that using the NB significantly alters the environment and may interfere with expected physiological responses to spaceflight that were identified previously and expected during long duration missions, thereby diminishing the scientific value of the RR animal model used to study biological effects of spaceflight.This mission is directed research by Space Biology and the plan was developed following extensive consultation and collaboration with the NASA Chief Veterinarian, Flight IACUC, the RR Science Working Group (SWG), welfare, and other science experts. This mission will provide reference data on animals' physiological systems and behavior within the Rodent Habitat to inform the NB's role in 1) supporting rodent health and welfare in space, and 2) to inform how the NB affects key tissues and science disciplines in both the space and ground environment. To accomplish mission objectives, adult wild-type female mice will be flown to the International Space Station (ISS) using RR Hardware. Groups of mice provided with the NB will be compared to those without the NB in the Rodent Habitat in both Flight and Ground. Comparison of Flight groups (FLT) to their respective Habitat Ground Control (HGC) groups will reveal if and how the NB affects major physiological systems relevant to the discipline of Space Biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Rodent Research Standard Housing Mission (RR-26) is a critical multi-disciplinary experiment designed to assess and improve translation of NASA rodent spaceflight studies with the goal of informing crew health on long duration missions. This will be accomplished b providing evidence to: (1) help define the role of a physical structure, or Nestbox, in supporting aspects of rodent. health and welfare in space, and determine whether provision of the Nestbox limits the study of certain scientific disciplines.
Task Progress:	This project is a continuation of "Rodent Research Standard Housing Mission: Multidisciplinary Approach to Understanding Spaceflight Responses" under previous PI, Dr. April Ronca. For earlier progress reporting, please see the record under the the previous PI (https://).
Bibliography Type:	Description: (Last Updated:)